

CANSYSPLUS

MONITORING IS OUR PASSION

VEHICLE TRACKING SYSTEM



GPS/GPRS Based
Web Based Interface
Google Earth & Google Map
Sensor Inputs
Relay Outputs

Dealer Distributor
All Over India
Solicited

OEM Manufacturing
Design & Firmware
Technology Assistance

Also Deal In

GSM/GPRS Modem
Embedded Systems
Handheld & Mobile Software
Data Loggers
Serial & Ethernet Devices
Wireless Devices

CANSYS PLUS SOLUTIONS PVT LTD

Works:

SCF 67, Sector-7, (1st & 2nd floor)
HUDA Complex, Faridabad, Haryana
91-129-4080851
9811205879, 9811905879

Head Office:

A-35, Nand Bhuvan Industrial Estate
Mahakali Caves Road, Andheri(E)
Mumbai, Maharashtra
91-22-67100700
9892100255, 9324175961

Distributors:

N.C.R	9810548584
New Delhi	9910977662
Madurai	9443351161
Nagpur	9325372162
Mumbai	9892100255
Hyderabad	9393322889
Kolkata	9832085816, 9432581244
Punjab	9878006677

Q1. How does the regenerative braking work and how the kinetic energy of braking is changed into electricity and reverted to the power source?

Deepak Kumar Saha
Mankundu

A1. A regenerative brake is a mechanism that reduces a vehicle's speed by converting some of its kinetic energy into another useful form of energy. This captured energy is then stored for future use or fed back into a power system for use. The storage device could be a battery, a bank of capacitors, compressed-air tank or a flywheel.

In electric railway wagons, the regenerated electricity is fed back into the supply system. In electric and hybrid vehicles, the energy is stored in a battery or a bank of capacitors for use later.

Regenerative braking utilises the fact that an electric motor can also act as a generator. The vehicle's electric traction motor is operated as a generator during braking and its output is supplied to an electrical load. It is the transfer of energy to the load which provides the braking effect.

This technique was implemented by American Motors Corporation (AMC) and Gulton Industries in a concept car called 'Amitron.' They used a piggyback system of two 11kg nickel-cadmium batteries and two 34 kg lithium batteries to power the car for 240 km when traveling at a speed of 80 kmph. The car's lithium batteries were designed for sustained speeds. During acceleration, the nickel-cadmium batteries would cut in briefly to boost the Amitron from a standstill to 80 kmph in 20 seconds. An energy regeneration brake system would automatically switch the drive motors to generators as the car slowed so that the batteries could recharge, thus increasing the range of the car.

Another example is that of electric railway wagon operation. During braking, the traction motor connections are altered to turn them into electrical gen-

erators. The motor fields are connected across the main traction generator and the motor armatures are connected across the load. The motor traction generator now excites the motor fields. The rolling locomotive or multiple unit wheels turn the motor armatures, and the motors act as generators, sending the generated current back into the supply. The current flows through the motor armatures during braking in one direction and it is opposite to that during motoring. Therefore the motor exerts torque in a direction that is opposite from the rolling direction.

Braking effort is proportional to the product of the magnetic strength of the field windings, times that of the armature windings.

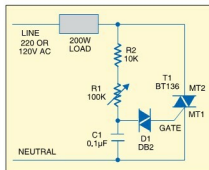
Savings of up to 17 per cent may be achieved. There is also less wear on friction braking components.

Q2. What is the difference between analogue and digital fan regulators?

Anuja Wadakar
Nagpur

A2. Conventional fan regulators work on the principle of auto-transformers. Another way is to use an electronic circuit with devices like triac and diac to control the speed. It may not be right to call it a digital fan regulator circuit.

Speed can be controlled by varying frequency or voltage. Varying frequency is not economical, so practically voltage-varying methods are used. Conventional fan regulators change the voltage by creating a voltage drop across the series resistance. Whereas,



Circuit of a fan regulator